Glossary

dating

Placing a sample or event in time.

chronology

The temporal ordering of evidence and events.

relative age

An older-or-younger relationship to other evidence.

numerical age

An age expressed numerically, for example in years.

calendar age

An age placed on a calendar timescale.

temporal resolution

The time interval resolved by a record.

uncertainty

The limited certainty of a measurement or inference.

radiocarbon dating

Dating based on radiocarbon in a sample.

calibration

Relating radiocarbon ages to calendar ages using a calibration curve.

half-life

The time for half the atoms of a radioactive isotope to decay.

stratigraphy

The study of strata and their relationships.

luminescence dating

Dating time since mineral signals were reset by light or heat.

palaeomagnetism

Records of past geomagnetic fields preserved in materials.

dendrochronology

The study of change using calendar-dated tree rings.

proxy

Evidence used to infer an environmental variable indirectly.

paleoclimatology

The study of past climates.

varve

A sedimentary layer verified to represent one year.

vascular cambium

Meristem producing secondary xylem and phloem.

xylem

Tissue involved in water transport and support.

phloem

Tissue transporting products of photosynthesis.

tracheid

A principal conducting and supporting cell of conifer wood.

vessel

A water-conducting structure composed of vessel elements.

ray

Radially oriented tissue in wood.

earlywood

Wood formed in the earlier part of a growth increment.

latewood

Wood formed in the later part of a growth increment.

tree-ring boundary

The boundary between adjacent annual rings.

growth ring

A ring-like increment associated with a growth cycle.

sapwood

The outer, physiologically active portion of wood.

heartwood

Inner wood transformed from sapwood, usually no longer conducting water.

false ring

An intra-annual feature that resembles an annual boundary.

locally absent ring

An annual ring absent from part of a stem.

crossdating

Matching ring patterns to establish and verify ring dates.

limiting factor

An environmental or physiological factor limiting growth.

reference chronology

A dated chronology used as a reference for crossdating.

floating chronology

An internally dated series not yet anchored to calendar years.

ring width

The radial width of an annual ring.

standardization

Processing growth trends and scales for the purpose of comparison.

ring-width index

Ring width expressed relative to an estimated growth trend.

dendroarchaeology

The study of past human activity using tree rings.

felling year

The year in which a tree was cut down.

dendroecology

The study of ecological processes using tree rings.

growth release

An increase in growth, for example after reduced competition.

disturbance

An event or process altering forest structure or growth.

dendroclimatology

The study of climate variability using tree rings.

correlation

A statistical association between variables.

regression

Modelling a relationship between variables.

validation

Checking model performance using independent information.

stable isotope

An isotope that does not undergo radioactive decay.

cell

A basic unit of biological structure and function.

tissue

An organized group of cells with related functions.

organ

A functional structure composed of several tissues.

meristem

A region of dividing cells contributing to plant growth.

primary growth

Growth increasing the length of stems or roots.

secondary growth

Growth increasing thickness through lateral meristem activity.

cell differentiation

Development of specialized cell structure and function.

cell enlargement

Increase in the size of a developing cell.

secondary cell wall

Wall deposited inside the primary wall, often after cell expansion.

lignification

Deposition of lignin in cell walls.

parenchyma cell

A cell involved in storage and metabolism, including living cells in wood.

ring-porous wood

Wood with large vessels concentrated near the start of a growth ring.

diffuse-porous wood

Wood with vessels more evenly distributed in size through a growth ring.

transverse section

A section approximately perpendicular to the stem axis.

radial section

A longitudinal section along a stem radius.

tangential section

A longitudinal section approximately tangential to the growth rings.

photosynthesis

Production of organic compounds using light energy and carbon dioxide.

respiration

Metabolic processes releasing energy from organic compounds.

transpiration

Loss of water vapour from plants, especially through stomata.

stoma

An opening regulating gas exchange and water loss.

carbon allocation

Partitioning acquired carbon among growth, storage and maintenance.

phenology

The seasonal timing of events such as leaf-out and dormancy.

dormancy

A state of temporarily suspended or reduced development.

physiology

The study of biological functions and processes.

ecology

The study of relationships among organisms and their environment.

population

Individuals of the same species in a defined area.

community

Populations of different species occurring together.

ecosystem

Organisms and the abiotic environment interacting as a system.

competition

Interactions involving access to limited resources.

canopy gap

An opening in the forest canopy following events such as tree death.

ecological succession

Change in community composition and structure through time.

biotic and abiotic factors

Biological influences such as competition, and physical influences such as temperature and moisture.